Long COVID Medical Documentation

Prepared for RTHM Pharmacy — medication support documentation | Prepared: July 18, 2026

PATIENT NAME
Hannah Munguia-Flores

DATE OF BIRTH
August 18, 1995 (age 30)

MRN (BWF)
10161495501

BLOOD TYPE
A Negative

PCP / ORDERING PROVIDER
Kristen Elaine Medley, MD, MPA

PRIMARY LAB
Brigham and Women's Faulkner Hospital

OCCUPATION
MIT Graduate Student (indefinite medical leave)

PRIMARY DIAGNOSIS
Long COVID / Post-Acute Sequelae of SARS-CoV-2

1. PURPOSE OF THIS DOCUMENT

This document provides a comprehensive summary of Hannah Munguia-Flores's medical history, laboratory findings, biometric data, current diagnoses, and treatment history to support RTHM Pharmacy's medication verification and dispensing process for Long COVID–related therapies.

Hannah has been living with Long COVID since approximately late 2022. Her illness is well-characterized through serial lab work, two wearable devices (WHOOP + Visible), and clinical follow-up with her PCP and naturopathic provider. This document was compiled from her full medical record by the clinical AI system supporting her care.

2. ACTIVE DIAGNOSES & WORKING CLINICAL PHENOTYPE

DIAGNOSIS	STATUS	EVIDENCE BASE
Post-Acute Sequelae of SARS-CoV-2 (Long COVID)	ACTIVE — PRIMARY	Clinical diagnosis; onset late 2022; biometric + self-report longitudinal data
EBV Reactivation / Viral Persistence	CONFIRMED LAB	EBV Early Antigen IgM reactive (Jun 19, 2025); positive mono screen Oct 2022; prior latent EBV
Post-Exertional Malaise (PEM)	ACTIVE	Data-confirmed 3-day lag; 774-day biometric dataset; pace_lag3 r = −0.74
Dysautonomia / POTS	ACTIVE	Resting tachycardia 103 bpm (Apr 2024); chronic hyponatremia; low HRV (17–49 ms); LMNT dependency
Hypothalamic-Pituitary Axis Dysregulation	CONFIRMED LAB	LH 0.01, FSH 0.46, Estradiol <10, Prolactin 22.48 (Jun 2025) — central, not ovarian
CIRS / Mold Illness (Chronic Inflammatory Response Syndrome)	UNDER TREATMENT	ERMI 25.8 (Red Zone), HERTSMI-2 20; all 14 IgG+IgE mycotoxin markers elevated (Dec 2024); moved Feb 2025
GI Mucosal Injury / Patchy Healing Ulcers	CONFIRMED PROCEDURE	Colonoscopy Sep 2025 (San Diego): patchy healing ulcers; celiac ruled out by biopsy; EGD clear
Tick-Borne Co-Infections	UNDER EVALUATION	Babesia duncani IgG+, Bartonella vinsonii IgG+, Anaplasma HGA IgM titer 40 (IgeneX, Mar 2024)
MCAS / Histamine Intolerance	ACTIVE	Clinical diagnosis; Xolair (omalizumab) ongoing; HistaQuel + low-histamine diet
Iron Deficiency Anemia (improving)	Improving	Ferritin 5–11 (2024) → now elevated at 191 (Jun 2026; inflammatory ferritin, not iron overload)
ADHD	ACTIVE	Established diagnosis; Adderall PRN (biometric correlation with function: +0.67)

3. CURRENT MEDICATIONS & SUPPLEMENTS

MEDICATION / SUPPLEMENT	DOSE / FREQUENCY	INDICATION	STATUS
Adderall (amphetamine salts)	IR, PRN	ADHD	Active
Low-Dose Naltrexone (LDN)	3 mg (tapered from 6 mg)	Neuroinflammation / Long COVID	Active — tapering
Xolair (omalizumab)	Monthly injection	MCAS / IgE-mediated reactions	Active (last: Jun 15, 2026)
Nystatin	Per protocol	Antifungal — CIRS/mold protocol	Active
Itraconazole	Per protocol	Antifungal — CIRS/mold protocol	Active
BCP-157 (oral peptide)	Per protocol	Gut healing / tissue repair	Started May 21, 2026
Rapamycin	Periodic dosing	mTOR modulation	Started May 26, 2026
Biocidin	Per protocol	Antimicrobial botanical — CIRS	Active
SIM01 Probiotic (G-NiiB Immunity Elite)	Nightly	Gut microbiome restoration (RECOVERY trial formulation)	Active
Zyrtec (cetirizine)	PRN	MCAS / histamine events	Active
LMNT electrolytes	Daily	Dysautonomia / POTS electrolyte support	Active
Vitamin D	Per label	Supplementation (D 49 ng/mL, optimal)	Active
Melatonin	PRN	Sleep support	PRN
Prozac (fluoxetine) 20 mg	Tapered → discontinued	Mood	Discontinued June 2026

Pending Prescription (Primary Unmet Need):
Valacyclovir 1g TID × 3–6 months for confirmed EBV reactivation. Letters requesting this prescription were submitted to Dr. Medley (PCP) and her naturopathic doctor in June 2026. As of this document, not yet prescribed.

4. LABORATORY RESULTS

4a. Most Recent Labs — June 25, 2026 (Brigham & Women's Faulkner Hospital)

Provider: Kristen Elaine Medley, MD, MPA | Collection: Jun 25, 2026 at 10:37 AM | CLIA #22D0663738

Abnormal Values (Jun 25, 2026):

- **Ferritin 191 ug/L** (ref 30–150) — HIGH. Iron/TIBC/transferrin saturation all normal → pattern consistent with *inflammatory ferritin elevation*, common in Long COVID. Not iron overload.
- **Triglycerides 355 mg/dL** (ref ≤150) — markedly HIGH. LDL/HDL/cholesterol all favorable. HbA1c and TSH normal. Likely dysregulated lipid metabolism in inflammatory/post-viral context.

PANEL	TEST	RESULT	UNITS	REFERENCE	FLAG
Hepatic Panel (LFTs)	AST	20	U/L	10–50	Normal
	ALT	10	U/L	10–50	Normal
	Alkaline Phosphatase	40	U/L	40–130	Normal
	Bilirubin, Total	0.2	mg/dL	0.0–1.2	Normal
	Bilirubin, Direct	0.1	mg/dL	0.0–0.3	Normal
	Total Protein	7.6	g/dL	6.4–8.3	Normal
	Albumin	3.9	g/dL	3.5–5.2	Normal
	Globulin	3.7	g/dL	1.9–4.1	Normal
Iron Studies	Ferritin	**191**	ug/L	30–150	**HIGH ⚠**
	Iron	92	ug/dL	28–170	Normal
	TIBC	347	ug/dL	220–460	Normal
	Transferrin Saturation	27	%	14–50	Normal
Vitamin D	25-OH Vitamin D, Total	49	ng/mL	20–50	Normal (optimal)
Thyroid	TSH	3.32	uIU/mL	0.40–5.00	Normal
Basic Metabolic Panel	Sodium	136	mmol/L	136–145	**Low-normal**
	Potassium	4.1	mmol/L	3.4–5.1	Normal
	Chloride	101	mmol/L	98–107	Normal
	CO2	27	mmol/L	20–31	Normal
	BUN	10	mg/dL	6–23	Normal
	Creatinine	0.73	mg/dL	0.50–1.00	Normal
	Glucose	84	mg/dL	70–99	Normal
	Calcium	9.4	mg/dL	8.5–10.5	Normal
	eGFR	113	mL/min/1.73m²	>59	Normal (excellent)

	Anion Gap	8	mmol/L	3–17	Normal
CBC	WBC	8.43	K/uL	4.00–11.00	Normal
	Hemoglobin	12.8	g/dL	12.0–16.0	Normal
	Hematocrit	38	%	36.0–46.0	Normal
	MCV	90	fL	80.0–100.0	Normal
	Platelets	324	K/uL	150–450	Normal
	RDW-CV	12	%	11.5–14.5	Normal (improved from 17.0 in 2024)
	RBC Morphology	Normal	—	—	Normal (was abnormal 2024)
Lipid Panel	Total Cholesterol	180	mg/dL	<200	Normal
	HDL	59	mg/dL	≥40	Normal
	LDL (calculated)	66	mg/dL	<130	Normal
	Cardiac Risk Ratio	3.1	ratio	0.0–5.0	Normal
	Triglycerides	**355**	mg/dL	≤150	**HIGH** ⚠
Glycemic	HbA1c	5.2	%	4.3–5.6	Normal

4b. Prior Lab History — Key Findings (2022–2025)

DATE	TEST	RESULT	SIGNIFICANCE
Oct 2022	Mononucleosis Screen	**Positive**	Primary EBV infection — onset of LC trajectory
Mar 2024	Ferritin	**5 ng/mL (ref 13–150)**	Severe iron deficiency
Mar 2024	RDW	**16.5–17.0%**	Worsening iron-deficiency anemia
Mar 2024	hs-CRP	**4.1 mg/dL (ref <3.0)**	Active systemic inflammation
Mar 2024	Babesia duncani IgG (IgeneX)	**Positive**	Past/chronic tick-borne Babesia
Mar 2024	Bartonella vinsonii IgG (IgeneX)	**Positive**	Past/chronic tick-borne Bartonella
Mar 2024	Anaplasma HGA IgM (IgeneX)	**Titer 40**	Active/recent Anaplasma signal
Mar 2024	Mycotoxin panel (RealTime, urine)	**Trichothecene PRESENT**	Stachybotrys mold exposure confirmed
Mar 2024	tTG-IgA (celiac serology)	**Positive**	Positive; later negative × biopsy — resolved
Oct 2024	IgA	**414 mg/dL (ref 47–310)**	Mucosal immune activation — mold/gut

Oct 2024	HbA1c	**5.8% (prediabetes range)**	Improved to 5.2% by Jun 2026
Dec 2024	MyMycoLab IgG+IgE (blood)	**ALL 14 markers elevated**	Systemic mold immune response (CIRS)
Dec 2024	ERMI (apartment, 4 Ellsworth Ave Cambridge)	**25.8 / HERTSMI-2 = 20**	Red Zone — Stachybotrys, Aspergillus, Penicillium
Jun 19, 2025	EBV Early Antigen IgM	**0.51 — Reactive**	Active EBV reactivation (not latent)
Jun 19, 2025	LH / FSH / Estradiol	**0.01 / 0.46 / <10**	Severe HPG axis suppression (central)
Jun 19, 2025	Prolactin	**22.48 ng/mL (ref 3-8)**	Elevated — suppressing GnRH axis
Jun 15, 2025	Cortisol	**25 µg/dL (ref 3.7–19.4)**	HPA hyperactivation — chronic stress response
Jun 15, 2025	Sodium	**133.4 mEq/L**	Dysautonomia-consistent; improved to 136 Jun 2026
Jun 19, 2025	Anti-cardiolipin IgM	**24.47 APL (ref >15 = Pos)**	Autoimmune/hypercoagulability signal
Sep 8, 2025	Stool occult blood	**POSITIVE (24 µg/g, ref <20)**	GI mucosal erosion confirmed
Sep 2025	Colonoscopy + EGD	**Patchy healing ulcers (colonoscopy)**	Celiac ruled out by biopsy; upper GI clear

5. BIOMETRIC OVERVIEW (WHOOP + VISIBLE)

Hannah has been tracked continuously on two wearable devices, providing an objective longitudinal dataset that supplements and contextualizes her lab work and clinical findings.

5a. WHOOP — Recovery, HRV & Sleep (90-Day Window: Mar 28–Jun 26, 2026)

METRIC	SUMMARY	CLINICAL CONTEXT
Recovery Score	~40% Green (≥70%), ~40% Yellow (34–69%), ~20% Red (<34%). Peak: 98% (May 31). Floor: 18% (May 30).	Predominantly yellow-zone — body consistently under-recovered. Sporadic green runs, no sustained streak yet.
HRV (rMSSD)	Range: 17–49 ms. Baseline: 26–37 ms. Peak: 49.4 ms. Nadir: 17.1 ms.	Well below expected for a 30-year-old female (normal: 45–65 ms). Reflects chronic autonomic nervous system suppression consistent with LC dysautonomia.
Sleep Performance	Highly variable: 7–85%. Frequent nights below 50%. Extreme lows: 14% (May 30), 19% (Jun 1).	Sleep architecture severely disrupted. Poor sleep nights precede PEM crashes. Sleep is the primary lever in her pacing protocol.
Daily Strain	Predominantly sedentary (0.1–4.0 most days). Spikes to 8–15 on exertion days → PEM 2–3 days later.	Near-baseline strain required for stability. Exertion spikes are strongly predictive of crashes.
Dataset size	2,292 days total WHOOP data backfilled. Analysis covers full history.	One of the largest individual LC biometric datasets. Statistical finding: pace_lag3 r = −0.74 (strongest crash predictor).

5b. Visible App — Stability Score & PacePoints (Apr 15–Jun 24, 2026)

PERIOD	AVG STABILITY SCORE (1–5)	AVG PACEPOINTS	TREND
April 15–30	2.9	~15	Variable, above protocol
May 1–15	3.3	~7.5	Improving adherence
May 16–31	3.5	~7.8	Good adherence
June 1–15	3.3	~8.0	Stable with minor spikes
June 16–24	4.0	~6.9	**Best sustained stretch — below protocol targets**

> **Pacing Protocol Targets (Repair Spectrum Framework):**
> Based on 774-day analysis: ≤8 PacePoints on yellow/first-green days; ≤12 on second green; ≤14 on third+ green. The June 16–24 period represents the strongest pacing adherence on record.

6. CURRENT FUNCTIONAL STATUS

DOMAIN	STATUS
Daily activity	Predominantly housebound. Light activities (sewing, short kitchen trips) on good days.
Outing tolerance	~1–2 hour outings with significant accommodations (pre-rest, seated rest after, sunglasses). Newport walking tour June 20, 2026 — a notable milestone.
Cognitive load	Brain fog documented. ADHD overlay amplifies sensory processing demands — 5–10 activities stack simultaneously against depleted energy when leaving home.
Occupational	On indefinite medical leave from MIT graduate program. TA role recommended as fully remote and async only.
Estimated capacity	~15–25% functional capacity on best days; ~5–10% on bad days.
Self-report check-ins	70 check-ins (Apr 26–Jun 24, 2026): 7% Good / 53% Mixed / 40% Bad. PEM rated severe on ~40% of check-in days.
Stellate Ganglion Block	SGB #1 May 29; SGB #2 June 12 (temp spike 99.5°F, then improvement). Used for dysautonomia/POTS management.

7. TREATMENT RATIONALE FOR REQUESTED MEDICATIONS

The following outlines the clinical justification for medications Hannah is seeking through RTHM Pharmacy, based on her confirmed diagnoses and current evidence base for Long COVID treatment.

Valacyclovir 1g TID × 3–6 months

Indication: EBV reactivation in Long COVID

Lab confirmation: EBV Early Antigen IgM reactive (0.51 ratio, Jun 19, 2025); historical positive mono screen (Oct 2022)

Evidence: Iwasaki Lab (Yale) — EBV reactivation is one of four documented root mechanisms in Long COVID. Valacyclovir is preferred over acyclovir (~55% vs ~20% oral bioavailability). Standard LC suppression dosing is 1g TID.

Evidence: Komaroff & Lipkin, PNAS 2023 — Viral persistence (including EBV) as a driver of LC pathophysiology.

Clinical rationale: Every EBV reactivation episode carries risk of re-injuring a GI mucosa already in recovery from mold-driven damage (colonoscopy-confirmed healing ulcers). Antiviral therapy is the single largest unmet treatment gap in Hannah's current protocol.

Drug interactions: None with current medication list. No nephrotoxicity concern (creatinine 0.73, eGFR 113 — excellent renal function).

Status: Not yet prescribed. Requests submitted to PCP and ND in June 2026.

Fluvoxamine (consider after valacyclovir is established)

Indication: Long COVID fatigue / neuroinflammation

Evidence: REVIVE-TOGETHER RCT (Reis et al., Annals of Internal Medicine, 2026) — n=399, stopped early. Fluvoxamine significantly reduced Long COVID fatigue vs. placebo (FSS −0.58 at day 90, 99% posterior probability). Benefit declined post-cessation, suggesting maintenance is needed.

Mechanism: Sigma-1 receptor agonism → reduces gut spike protein-driven neuroinflammation. Relevant to Hannah's gut involvement + EBV reactivation pattern.

Interaction note: CYP1A2/2D6 inhibitor — review full drug list for interactions before starting. Currently no conflicting agents on board.

Low-Dose Aripiprazole 0.25–2mg (if brain fog/dysautonomia remain limiting)

Indication: Brain fog / dysautonomia in Long COVID

Evidence: Stanford Long COVID Clinic retrospective (2025) — 50 LC patients; 22% had ≥30% symptom reduction. RTHM and PLRC include in their 2026 treatment guidance.

Mechanism at low dose: Restores dopaminergic tone; suppresses neuroinflammation. Completely distinct from psychiatric dosing (standard: 10–30mg). At 0.25–2mg it functions as a dopamine partial agonist, not a neuroleptic.

Relevance: Brain fog and dysautonomia are among Hannah's most limiting persistent symptoms.

8. GENETIC & PHARMACOGENOMIC NOTES (23ANDME)

FINDING	CLINICAL RELEVANCE FOR PRESCRIBING
SLCO1B1 — Decreased function	Avoid simvastatin. If lipid management needed: use rosuvastatin or pravastatin.
Celiac genetic risk (HLA variants)	Increased susceptibility confirmed. Clinically resolved by colonoscopy biopsy — not true celiac.
Hashimoto's + Lupus predisposition	Monitor thyroid (TSH 3.32 currently normal). Autoimmune vigilance warranted.
NAFLD predisposition	Monitor LFTs long-term (currently all normal).
Lactase persistent	Lactose intolerance not a GI driver.
MTHFR	Not yet tested. Recommend C677T/A1298C genotyping given mold illness + methylation pathway relevance.
G6PD	Testing ordered twice (Jul + Oct 2024); result not in file. Required for methylene blue safety if ever considered.
HFE (hereditary hemochromatosis)	No variants. Not the cause of prior low ferritin.
Factor V Leiden / hereditary thrombophilia	Not detected. Anticardiolipin IgM positive (Jun 2025) — repeat at 12-week interval for APS criteria.

9. CLINICAL TRAJECTORY & SUMMARY ASSESSMENT

What is improving

- Electrolyte normalization: Sodium 133 → 136 mEq/L, CO_2 normalized — partial dysautonomia stabilization
- Iron: Ferritin normalized from 5 → 191 (now elevated; inflammatory, not deficiency)
- CBC: RDW normalized (17.0 → 12.0%), anemia resolved, RBC morphology normal
- HbA1c: Improved from 5.8% (prediabetes) to 5.2% (normal)
- Visible Stability Score: Trending upward (2.9 Apr → 4.0 June avg)
- Pacing discipline: PacePoints trending down — best adherence in June 2026
- Post-SGB June period showed subjective improvement

What remains persistently abnormal

- **EBV reactivation — active, untreated.** No antiviral on board. Single largest gap.
- HRV 26–37 ms vs expected 45–65 ms for age — chronic autonomic suppression
- 40% bad days on self-report; severe PEM on ~40% of check-in days
- Triglycerides 355 mg/dL — new finding (Jun 2026), cause unclear
- Ferritin 191 — inflammatory elevation, not resolved

> **Primary Unmet Clinical Need:**
> Antiviral therapy targeting active EBV reactivation (valacyclovir 1g TID × 3–6 months). This is the most evidence-aligned next step based on her confirmed lab findings, established Iwasaki Lab protocols, and her colonoscopy-confirmed GI mucosal vulnerability to EBV-driven flares. It has not yet been initiated due to prescription delays.

Document prepared by: Cleo — Clinical AI System, Spectator Health | Compilation date: July 18, 2026

Data sources: WHOOP biometric (2,292-day dataset), Visible app (Apr–Jun 2026), patient check-ins (Apr 26–Jun 24, 2026), Brigham & Women's Faulkner Hospital labs (Jun 25, 2026), prior labs 2022–2025, IgeneX panel, MyMycoLab, 23andMe, colonoscopy Sep 2025

PCP: Kristen Elaine Medley, MD, MPA | BWF MRN: 10161495501 | DOB: 8/18/1995

This document is a clinical data summary prepared to support care coordination. It is not a substitute for physician evaluation.